

ECEN 662: ESTIMATION & DETECTION THEORY

Prof. Tie Liu | Spring 2026 | Tuesday & Thursday 02:20 PM – 03:35 PM @ ETB 1003

OVERVIEW

This course covers the fundamentals of statistical inference, including both statistical principles and computational techniques such as variational inference and sampling methods. This course is the natural continuation of the graduate-level probability/random-process class and is designed to prepare students to take advanced machine learning classes.

REQUIREMENTS

- Weekly homework (75%):
 - **Theoretical** in nature
 - Occasionally **very** challenging
 - Discussions among peers are encouraged.
However, each student must submit his/her own **independent** work
- Term project (25%)
 - The project can be either a specific application, or some preliminary research on methodology/algorithm
 - The topic of the project must be **pre- approved** by the instructor

TEXTBOOKS

- E. L. Lehmann and George Casella: Theory of Point Estimation, Second Edition. Springer 1998
- E. L. Lehmann and Joseph P. Romano: Testing Statistical Hypotheses, Third Edition. Springer 2008
- Christian P. Robert and George Casella: Monte Carlo Statistical Methods, Second Edition. Springer 2004

TOPICS

- Minimum-variance unbiased estimation (MVUE)
- Maximum-likelihood estimation (MLE)
- Expectation-maximization (EM) algorithm
- Minimum-risk equivariant estimation (MREE)
- Bayes estimators
- Variational inference
- Empirical Bayes estimation
- Minimax estimators
- Uniformly most powerful (UMP) tests
- Transform and rejection sampling
- Monte Carlo integration
- Markov chain Monte Carlo (MCMC) sampling
- Diffusion Monte Carlo sampling